

What a Chance Constraint Does Not Promise

Measuring a demand-ceiling guarantee across thirty electricity supplies in eight countries, with India as the focal case

Swetank Kumar

Project Samanvay / Aethergrid

Abstract

Indian commercial demand charges are set by a single worst thirty-minute block in a month, which makes defending a demand ceiling a risk problem rather than a prediction problem. The standard applied response is to substitute a high forecast quantile into the capacity constraint and call the result a 95% guarantee. We measure what that guarantee is actually worth. On held-out meter data over 5,793 forecast origins, the per-step exceedance rate is 0.062 against a nominal 0.05 — marginally, the forecaster does exactly what it promised — while the probability of breaching the ceiling *somewhere* in the controller’s 16-hour planning window is 0.405. That is $6.5\times$ the per-step rate the forecaster delivers and $8.1\times$ the level the constraint appears to promise. Neither textbook substitute recovers it: the independence calculation $1 - (1 - \alpha)^H$ overstates the risk by $2.4\times$ because load errors are strongly autocorrelated, and the Boole/Bonferroni union bound — the only one of the two that is a valid bound — reaches 1 by $H = 17$ and is vacuous over the horizon the controller actually plans on. We then restore the level as a settable parameter with a copula-scenario MILP, tracking requested ε at rank correlation 0.957 for Rs 3,339 on a billing month, and report its exactness and resolution limits.

We test the rest of the pipeline by running one unchanged protocol over thirty electricity supplies in eight countries: eighteen buildings across eight usage classes from the Building Data Genome Project 2, six metered national or city demand series including the Delhi State Load Despatch Centre and ENTSO-E, and six Chinese provinces added late on an externally imposed constraint. Two findings follow. Distribution-free coverage degrades with aggregation level — mean 0.901 on buildings, 0.829 on metered system demand — and the adaptive layer, not split conformal, is what holds it. And forecast skill is uncorrelated with climate while available flexibility tracks it strongly, so deployment screening should ask how much load is weather-driven rather than how predictable it is; the null replicates on the Chinese panel it was not fitted to.

India is the focal case and returns a divided verdict: the most favourable load physics in the study and the worst calibration in it. The China arm’s most quotable result — that the closest structural analogue to Delhi returns the second-worst skill in the benchmark — is retracted in Section 8, because auditing that dataset against its own documentation showed a whole year of provincial demand to be 778 numbers reproducing 8,760 values to the precision the source is published at, which makes its skill column a statement about the data rather than about a forecaster. Five rows argue against the method, three of them outright negative, and all five are reported at the same size as the rest.

1 Introduction

An Indian commercial electricity bill has three levers: a time-of-day energy charge, a demand charge levied on the highest 30-minute average demand in the month, and power-factor penalties. The second dominates the risk. It is set by one block out of roughly 1,440 in a month, so a single bad afternoon prices all thirty days. A controller that reduces average consumption but occasionally overshoots can cost more than one that does nothing.

The standard applied response is a chance constraint: replace the forecast mean in the capacity constraint with a high quantile, typically the 95th, and treat the result as a 95% guarantee. Formally, with b_t the uncontrollable base load, u_t the controllable allocation, s_t solar generation and D the ceiling,

$$\hat{b}_t^{q95} + u_t - \hat{s}_t^{q05} \leq D \quad \text{for each } t \text{ in the horizon.} \quad (1)$$

Equation (1) is a set of *marginal* statements. Each row says that at that timestep, considered alone, the realisation exceeds the bound about 5% of the time. The event that costs money is different: it is

$$\Pr \left[\max_{t \in \mathcal{W}} (b_t + u_t - s_t) > D \right], \quad (2)$$

a statement about a maximum over a window $\mathcal{W}$. That (1) does not imply (2) is elementary and well known; joint chance constraints, their intractability, and the standard routes around them — Boole/Bonferroni risk allocation and the scenario approach [Calafiore and Campi, 2006, Nemirovski and Shapiro, 2006] — are textbook material.

What is not established is how large the gap is on real building load, and how badly each of the standard substitutes misses it. Both ends matter practically: the first is how much a deployed controller is under-protected, and the second is how much headroom a careful designer gives away for nothing. We measure both, and find they are wrong in opposite directions by factors we can state.

Contributions.

1. A measured bracket on horizon risk for a demand ceiling on held-out meter data (Section 4): claimed 0.05, realised 0.405, independence calculation 0.983, union bound vacuous. The common formulation is under-protected by $8.1\times$ and the usual correction is $2.4\times$ too conservative.
2. A closed-loop demonstration that this risk level can be made an operator-set parameter rather than an emergent property of horizon length, with its exactness and resolution floor measured and its price stated (Section 5).
3. Evidence that conformal coverage degrades with aggregation level, from one protocol run over thirty supplies in eight countries, together with the measurement showing that the adaptive layer and not split conformal is what holds the level (Section 6).
4. A null separating forecast skill from available flexibility as independent deployment axes, replicated on a panel it was not fitted to (Section 7).
5. A provenance audit of a dataset added to the study under an external constraint, which disqualifies that arm’s most favourable result and is reported as such (Section 8).

Where the novelty is, and where it is not. We state this precisely, because a novelty claim that cannot survive a literature search is worth less than no claim. *Not novel:* chance-constrained model predictive control for buildings; joint chance constraints and their standard treatments, including Boole/Bonferroni risk allocation and the scenario approach [Calafiore and Campi, 2006, Nemirovski and Shapiro, 2006]; quantile gradient boosting [Ke et al., 2017]; conformalized quantile regression [Romano et al., 2019] and adaptive conformal inference [Gibbs and Candès, 2021]; decision-focused learning [Elmachtoub and Grigas, 2022, Donti et al., 2017, Berthet et al., 2020]; and cross-building transfer learning, which is an active area operating on this very corpus [Miller et al., 2020, Nweye et al., 2023]. We invented none of it and do not claim to.

Novel: the instrument, and what it measures. We are not aware of prior work that (i) brackets the marginal-versus-horizon risk gap for a demand ceiling on real meter data with *both* ends reported, so that the common formulation and the textbook correction are shown to be wrong in opposite directions by measured factors; (ii) reports that distribution-free coverage degrades systematically with aggregation level, holding at building scale and failing at system scale, with the adaptive remedy measured over a walk-forward year; (iii) separates forecast skill from available flexibility as empirically independent deployment criteria, with the expected relationship reported as a null and replicated; or (iv) benchmarks a tariff-native demand controller across eight countries with India as the focal case and the Indian result reported as the weakest one. The novelty is empirical and integrative rather than algorithmic. That is a narrower claim than a new method, and it is one that holds.

2 The instrument

Six components, each independently swappable, with the bill engine as the single source of truth for every reported figure:

$$\begin{aligned} \text{tariff} \rightarrow \text{quantile forecast} \rightarrow \text{conformal calibration} \rightarrow \text{chance-constrained MILP} \\ \rightarrow \text{RC simulator} \rightarrow \text{bill engine.} \end{aligned}$$

The forecaster is a quantile LightGBM at five levels over a 64-step (16-hour) horizon at 15-minute resolution, wrapped in split and adaptive conformal calibration. The controller is a deterministic MILP

solved with HiGHS; no language model touches a money decision. The evaluation discipline is that a number reaches a table only if a script produced it, and a single entry point regenerates every figure and every table in this paper from a clean checkout.

This matters for the measurements because each one requires pinning everything and moving one thing. Section 4 pins the plant and varies only the risk formulation. Section 6 pins the protocol and varies only the series. Section 7 pins the model and varies the site. Section 8 pins all three and varies the country, which is the one we did not choose.

3 Data

Five public sources, all fetched by script rather than vendored, and named in full because the benchmark’s credibility rests on them.

Building Data Genome Project 2 [Miller et al., 2020]. 1,636 metered non-residential buildings across nineteen sites, of which 1,578 carry an electricity meter, hourly with matched weather, two years. It is the only harmonised multi-country building corpus in existence, which is why it carries the controlled comparison: the same physical quantity, the same metering convention, the same period. It contains *no Indian buildings*, and this is the single most consequential gap in the study — Section 10.

Delhi State Load Despatch Centre. Fifteen-minute metered demand for the city of Delhi, disaggregated across five distribution companies (BRPL, BYPL, NDPL, NDMC and MES) plus the total, two complete years at full coverage. This is the only series in the entire study at *native* fifteen-minute resolution — the cadence the controller runs at and the cadence on which Indian demand charges are assessed. Every other series is interpolated up from hourly.

ENTSO-E transmission data, via Open Power System Data. Hourly national load for the United Kingdom, Ireland, Germany, France and Spain, drawn from the European transmission system operators’ transparency platform.

Chinese provincial demand [Wu and Kan, 2023]. Hourly load for the 31 provinces of mainland China, 2018, from which six are drawn. This source was added after the study was otherwise complete, and it is the only tier-2 series that is constructed rather than metered. Section 8 audits it.

ERA5 reanalysis, via the Open-Meteo archive. Hourly temperature, dew point and cloud cover at each location, used identically for every series.

Windows. The building tier and the European national tier share 2016–2017. The Delhi archive covers 2011–2012, so the same protocol is shifted back five years with its shape preserved. The Chinese series is a single calendar year, 2018. In every row of the study the test month is June, and the headline metric is skill against each series’ own seasonal-naïve baseline: a ratio computed inside one series’ own data and its own period, which is what makes rows with different units and different years comparable at all. Absolute losses are not comparable across groups and are never compared.

Selection. One rule, stated in advance and applied uniformly: no district chilled-water meter, >97% coverage, median load 40–800 kW, positive load–temperature correlation; ties within a (site, usage) cell broken by highest p_{99} /median. Thirty supplies pass. Two facts about the rule were discovered by applying it rather than anticipated when writing it, and both are reported in Section 6.

4 The horizon gap

Table 1 and Figure 1 report, for a 13,759 m² office on held-out June 2017 data over 5,793 forecast origins, the per-step exceedance rate of the ceiling alongside the probability that the ceiling is breached at least once within a window of H steps.

Three readings of the last row are simultaneously true and together are the result.

H	hours	per-step $\hat{\alpha}$	realised	independent	Boole $H\hat{\alpha}$	copula
1	0.25	0.060	0.060	0.060	0.060	0.049
4	1.00	0.060	0.091	0.219	0.240	0.100
8	2.00	0.060	0.126	0.389	0.478	0.148
16	4.00	0.059	0.182	0.625	0.952	0.225
32	8.00	0.061	0.275	0.867	1.000	0.342
64	16.00	0.062	0.405	0.983	1.000	0.497

Table 1: Marginal versus horizon-level exceedance of the demand ceiling, held-out June 2017, 5,793 forecast origins. The per-step rate $\hat{\alpha}$ is well calibrated at every horizon against a nominal 0.05. The realised probability of breaching *somewhere* in the window is not that number. Neither substitute is usable: the independence calculation $1 - (1 - \hat{\alpha})^H$ overstates the risk by $2.4\times$ because load errors are strongly autocorrelated, and the Boole/Bonferroni union bound $\min(1, H\hat{\alpha})$ – the only one of the two that is a valid bound – saturates at 1 by $H = 17$ and is vacuous over the controller’s actual horizon.

Target	Controller	ε	Commit viol. (gap)	vs target	Breaches	Peak kVA	Bill Rs	Solve ms
tight	marginal $q95$	–	0.106	0.049	4	459.0	1,138,193	6
tight	scenario	0.05	0.097 (+0.047)	0.097	9	464.6	1,141,532	53
tight	scenario	0.10	0.114 (+0.014)	0.096	9	466.8	1,142,186	53
tight	scenario	0.20	0.156 (-0.044)	0.073	10	469.2	1,142,647	54
tight	scenario	0.35	0.219 (-0.131)	0.077	13	467.5	1,141,036	56
nominal	marginal $q95$	–	0.103	0.000	0	481.9	1,149,580	6
nominal	scenario	0.05	0.078 (+0.028)	0.000	0	480.5	1,148,576	52
nominal	scenario	0.10	0.097 (-0.003)	0.000	0	486.0	1,151,865	53
nominal	scenario	0.20	0.150 (-0.050)	0.000	0	480.1	1,148,071	53
nominal	scenario	0.35	0.208 (-0.142)	0.000	0	484.0	1,150,364	53

Table 2: Closed loop over one billing month. **Commit violation** is the acceptance metric: the fraction of horizons in which realised load cleared the ceiling the optimiser committed to, which is what ε is a statement about. The against-target column is the business metric and is not what the chance constraint promises, since the committed peak is a decision variable. Rank correlation 0.957, mean absolute gap 0.057, conservative at 5 of 8 levels; resolution floor $1/S = 0.025$.

The forecaster is not the problem. The per-step quantile is well calibrated at every horizon: $\hat{\alpha}$ holds between 0.059 and 0.062 against a nominal 0.05. Marginal calibration delivers exactly what it promised, which is why this is a result about a formulation rather than a bug report about a model.

The formulation is under-protected by a factor of eight. The probability of breaching somewhere in the 16-hour window is 0.405: $6.5\times$ the per-step rate the forecaster actually delivers, and $8.1\times$ the 0.05 that (1) appears to promise. A controller deployed on (1) and described to an operator as a 95% guarantee is running at roughly 60%.

Neither standard substitute recovers it, and they fail in opposite directions. This is the correction to the usual telling, and it matters. The quantity commonly quoted, $1 - (1 - \alpha)^H$, is the *independence* calculation, not a bound: it assumes the exceedance events are independent, which is the one thing load errors are not. It gives 0.983, which is $2.4\times$ the realised 0.405, because the fitted error process is heavily autocorrelated — lag-1 correlation 0.871, 0.578 at lag 4, decaying to 0.085 by lag 32 — so exceedances cluster rather than spread. The quantity that *is* a valid bound, Boole/Bonferroni’s $\min(1, H\hat{\alpha})$, makes no dependence assumption and pays for it: at $\hat{\alpha} \approx 0.06$ it reaches 1 by $H = 17$, four hours into a sixteen-hour horizon, after which it says nothing at all.

The practical consequence runs in both directions. A controller using (1) is materially under-protected. A controller sized off the independence figure is over-protected by more than a factor of two and pays for it in usable headroom, which is the quantity the entire economic argument rests on. A controller that insists on a valid bound over this horizon has no bound to use.

5 Restoring the guarantee as a dial

We fit a Student- t copula over the horizon and sample joint trajectories, which a scenario MILP then constrains with a CVaR surrogate on the probability mass permitted to violate. The copula reproduces

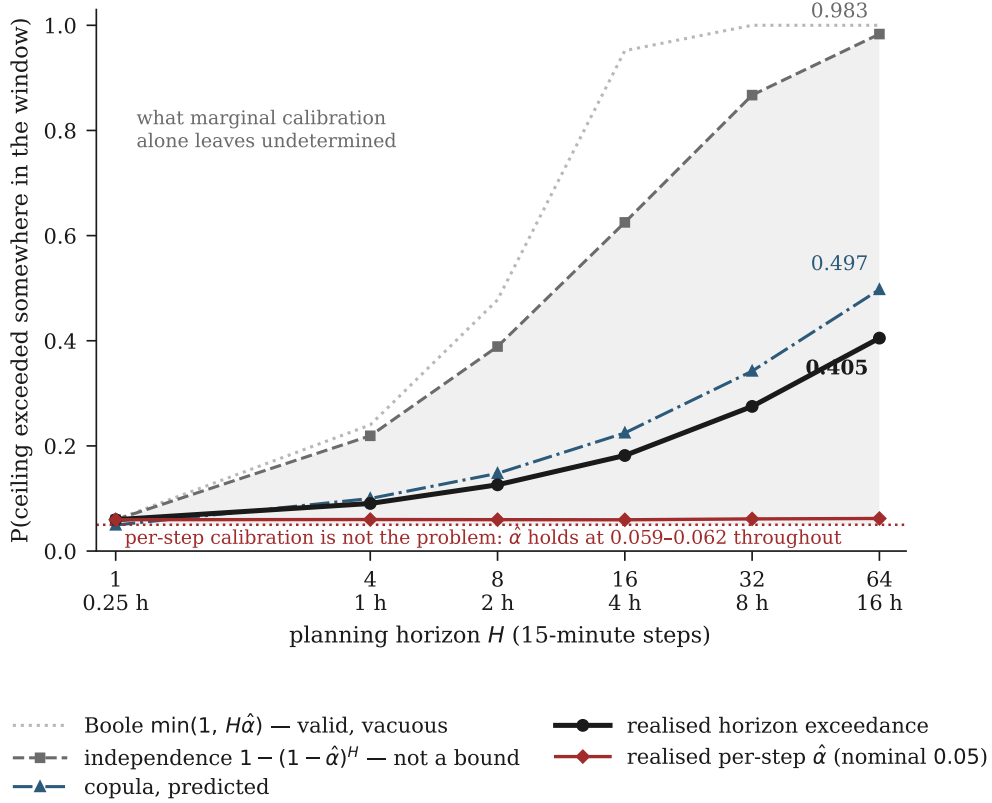


Figure 1: The bracket. Marginal calibration is a statement about each row of (1) separately; it constrains the probability of breaching somewhere in the window only to the shaded region, which at $H = 64$ spans 0.05 to 0.983. The per-step rate (bottom) is flat and well calibrated at every horizon, so nothing here is a broken forecaster. Held-out June 2017, one 13,759 m² office, 5,793 forecast origins.

the calibrated marginals to within 4.6% of mean interval width at 4,000 paths per origin, so it adds dependence and disturbs nothing upstream, and it predicts horizon exceedance far closer than either substitute: 0.497 against a realised 0.405 at $H = 64$, where (1) implies 0.05 and the independence figure is 0.983.

Table 2 is a closed-loop run over a real billing month at two demand targets and four risk levels. The acceptance metric is violation of the ceiling *the optimiser itself committed to at each origin*, because that is the event ε is a statement about. The operator’s monthly demand target is a business metric the chance constraint never promised: the committed peak is a decision variable, and an optimiser short of capacity may lift it, pay the demand charge, and honour its constraint exactly. Grading a chance constraint against a number it never promised is the usual way this test is made meaningless, so both columns are shown and only one is the acceptance criterion.

The result is that ε becomes a dial. Realised commit violation tracks the requested level with rank correlation 0.957 across the sweep, mean absolute gap 0.057, and lands conservative at five of eight levels. The formulation errs safe by construction: the CVaR surrogate admits a strict subset of the chance-constrained feasible set, so it defends the committed ceiling harder than asked.

What it does not do, stated in full. It is not exact. At the tightest setting, $\varepsilon = 0.05$ delivers 0.097 at the tight target, nearly twice the requested risk and on the unsafe side. There is a hard resolution floor at $1/S = 0.025$: with S scenarios the finest expressible risk level is one scenario’s worth of probability mass, below which the CVaR surrogate collapses onto the sample maximum and successive ε cease to be distinguishable. That is a property of sample average approximation rather than a tuning artefact, and a finer dial costs scenarios and solve time.

Three further caveats belong here rather than in a footnote. The copula’s degrees of freedom, $\nu = 7$, were selected on the validation block by matching predicted to realised horizon exceedance — which is the same quantity the copula is later judged on, so the agreement in the last column of Table 1 is not an independent test of the family. Base-load and PV scenarios are drawn independently, while cloud physically raises cooling load at the same moment it cuts PV output, so the true joint law has worse

Site	Country	CDD sh.	HVAC sh.	Seas. naive	Ours	Skill	Cov 90%
Phoenix, United States	United States	0.87	0.38	4.529	3.484	+0.231	0.884
Austin, United States	United States	0.75	0.13	1.352	1.061	+0.215	0.906
Washington DC, United States	United States	0.31	0.23	4.698	2.964	+0.369	0.900
Minneapolis, United States	United States	0.11	0.13	21.875	10.792	+0.507	0.916
Berkeley, United States	United States	0.05	0.26	5.855	3.033	+0.482	0.873
London, United Kingdom	United Kingdom	0.05	0.04	4.106	2.061	+0.498	0.910
Dublin, Ireland	Ireland	0.00	0.04	7.636	3.451	+0.548	0.918

Table 3: Climate arm: demographic held fixed at Education, climate and country varied across seven sites and three countries.

Site	Country	CDD sh.	HVAC sh.	Seas. naive	Ours	Skill	Cov 90%
Phoenix, United States	United States	0.87	0.24	6.213	2.419	+0.611	0.888
Washington DC, United States	United States	0.31	0.05	49.625	34.038	+0.314	0.824
Minneapolis, United States	United States	0.11	0.24	4.784	3.009	+0.371	0.914
London, United Kingdom	United Kingdom	0.05	0.01	13.893	20.322	-0.463	0.930
Dublin, Ireland	Ireland	0.00	0.03	4.150	2.020	+0.513	0.912

Table 4: Office arm: the same ladder in a second usage class, run as a control on the climate arm. The London office is the study’s worst row and is reported unchanged.

afternoons than these scenarios contain. And the horizon is sixteen hours while the demand charge bills a monthly maximum over roughly 2,880 intervals: a joint guarantee over the window is strictly stronger than a marginal one and strictly weaker than a monthly one. Extending it needs the copula fitted over a month-length horizon, which is a 2,880-square correlation matrix and needs structure rather than a raw estimate.

The honest summary is not that the guarantee is restored exactly. It is that a risk level which was previously an *emergent property of horizon length* — unstated, unsettable, and equal to 0.106 at the tight target under the incumbent marginal controller — is now a parameter with a monotone and conservative response, at a cost of Rs 3,339 on the month.

6 Calibration degrades with aggregation

One protocol — 15 months train, 2 months validate, 1 month test, temporal throughout, conformal calibration on the validation block — run unchanged over thirty supplies. Tables 3–7 give the five arms.

Table 8 and Figure 2 are the consistent finding. A nominal 90% interval holds at building level (mean 0.901; one row of eighteen below 0.85) and fails at system level (mean 0.829 metered, 0.824 reconstructed; five of six and two of six below 0.85 respectively). Delhi is the worst metered row in the study at 0.762 and Heilongjiang the worst row anywhere at 0.565.

This is consequential rather than curious, because the controller’s q_{95} is read from the same object. Too narrow an interval yields too narrow a demand ceiling, and the safety property degrades precisely on the load the project targets.

The mechanism. Split conformal guarantees coverage when the calibration and test blocks are exchangeable [Vovk et al., 2005, Romano et al., 2019]. A Delhi May is not exchangeable with a Delhi June, because the monsoon arrives between them and changes the relationship between temperature and load. This is an assumption failure, not a tuning failure, and it is diagnosable rather than speculative: our own audit partitions May into six disjoint calibration blocks and finds coverage varying from 0.710 to 0.861 with a block-bootstrap standard error of 0.031, failing every acceptance criterion stated in advance.¹

The remedy, at building scale. Figure 3 runs a walk-forward year. Split conformal ranges over 0.447–0.936 by month and spends 52% of the year inside a 0.85–0.95 band; adaptive conformal inference [Gibbs and Candès, 2021] ranges over 0.867–0.928 and spends 87%. Under a frozen model and a synthetic distribution shift the same ordering holds: split conformal settles at 0.770 coverage and never recovers

¹The bootstrap resamples thirty whole test days rather than 184,320 overlapping 15-minute forecasts, because those are not 184,320 independent observations. The naive standard error is 0.0007 and would pass any model ever built.

Usage class	Country	CDD sh.	HVAC sh.	Seas. naive	Ours	Skill	Cov 90%
Entertainment/public assembly	United States	0.31	0.31	4.193	2.038	+0.514	0.919
Healthcare	United States	0.31	0.21	2.907	1.702	+0.415	0.902
Retail	United States	0.31	0.25	7.681	4.734	+0.384	0.897
Warehouse/storage	United States	0.31	0.06	1.635	1.189	+0.273	0.942
Lodging/residential	United States	0.31	0.28	2.726	2.102	+0.229	0.901
Public services	United States	0.31	0.03	3.882	3.680	+0.052	0.887

Table 5: Demographic arm: climate held fixed at Washington DC, demographic varied across the eight usage classes that pass the selection rule at that site. Six rows appear here; Education and Office were also run in this arm and are shown in Tables 3 and 4 instead, since a building is one supply however many arms it appears in.

Series	Country	CDD sh.	HVAC sh.	Seas. naive	Ours	Skill	Cov 90%
India (Delhi)	India	0.89	–	72.499	65.437	+0.097	0.762
Spain	Spain	0.28	–	594.320	226.544	+0.619	0.813
France	France	0.08	–	942.571	342.809	+0.636	0.829
Germany	Germany	0.06	–	1344.698	386.465	+0.713	0.837
United Kingdom	United Kingdom	0.02	–	451.549	378.932	+0.161	0.822
Ireland	Ireland	0.00	–	29.812	16.624	+0.442	0.911

Table 6: National arm, metered: system-level demand in six countries. Delhi is 15-minute native resolution from the Delhi State Load Despatch Centre; the European series are ENTSO-E hourly via Open Power System Data.

within the run, ACI at 0.885. The layer with the finite-sample theorem is not the layer that delivers the level here, and that is worth stating plainly because it inverts the usual presentation order.

What remains open. Whether the adaptive layer restores the level at *system* scale — the tier where the failure is systematic, and the tier India sits in — is the obvious next experiment. We have the method and have not run it at that tier, and we decline to claim the result before we have.

Selection is regional, and we found that out the hard way. The rule’s final clause — positive load–temperature correlation — is a sanity check in Phoenix, where every candidate is cooling-dominated, and a site-destroying filter in Minneapolis or Dublin, where buildings consume electricity when it is cold. Seventy-seven buildings across nine sites pass every clause but that one. Separately, the two hottest sites in the corpus, the closest available analogues to an Indian metro climate, yield no admissible building at all for want of coverage in the split window. Both facts were discovered by applying the rule, not anticipated when writing it.

7 Skill and flexibility are independent axes

We expected forecast skill — measured with pinball loss, a strictly proper scoring rule [Gneiting and Raftery, 2007], against each series’ own seasonal-naive baseline — to increase with cooling load, on the reasoning that weather-driven demand is the learnable part. Table 9 and Figure 4 report the opposite: a null. Every correlation against skill is indistinguishable from zero and every leave-one-out range straddles or nearly straddles the sign line, which at these sample sizes is what a null result looks like. It replicates at $r = +0.054$ on the Chinese provincial panel, which was added afterwards and spans a wider climate range than the whole European set.

The controllable fraction, by contrast, tracks climate strongly ($r = +0.489$), running from 37.6% of the meter in Phoenix to 0.8% in London.

The deployment consequence is a screening rule. Do not ask whether a site’s load is predictable; ask how much of it is weather-driven. A Dublin office is highly predictable (skill +0.513) with roughly 3% controllable load, and there is nothing to sell there. Conflating the axes inverts the decision. For the target market this is favourable: Delhi demand correlates +0.59 with outdoor temperature against −0.64 for France, placing India at the useful end of the axis that matters.

Series	Country	Load-temp.	CDD sh.	HVAC sh.	Seas. naive	Ours	Skill	Cov 90%
Hainan (Haikou)	China	+0.10	0.97	–	131.650	93.878	+0.287	0.960
Guangdong (Guangzhou)	China	+0.37	0.85	–	2357.241	3097.316	-0.314	0.919
Shanghai	China	+0.07	0.44	–	445.551	352.338	+0.209	0.894
Beijing	China	-0.28	0.29	–	451.459	387.453	+0.142	0.905
Yunnan (Kunming)	China	-0.35	0.14	–	251.362	197.905	+0.213	0.703
Heilongjiang (Harbin)	China	-0.47	0.12	–	148.038	191.583	-0.294	0.565

Table 7: The China arm, added after the study was complete on a constraint imposed from outside it: six provinces spanning a wider climate range than the entire European panel, in a developing economy, at one calendar year and therefore a three-month training block rather than fifteen. Mean skill is +0.040 with 2 negative rows. **That column should not be read as a statement about the method**, for the reason established in Table 10: the series is constructed, not metered, and its construction pins our model and the baseline it is scored against to the same two numbers per day. Coverage does not run through the baseline and is readable; skill is not.

Population	n	mean cov.	worst	< 0.85
Buildings (BDG2), metered	18	0.901	0.824	1
System demand, metered	6	0.829	0.762	5
System demand, reconstructed	6	0.824	0.565	2

Table 8: Empirical coverage of the nominal 90% interval after conformal calibration. The guarantee very nearly holds on individual buildings and fails systematically on system-level demand. The third row is the replication: six Chinese provinces, a different country, a different year and a different data-generating process, reproduce the system-level failure to within 0.005 of the metered panel and contain the worst row in the study (Heilongjiang, 0.565).

8 The new country, and what it turned out to measure

The study was complete when a constraint was imposed from outside it: add a country. China was chosen because six provinces spanning Hainan to Heilongjiang cover a wider climate range than the entire European panel and do so in a developing economy, which is the comparison India actually needs. The protocol adaptation it forced is stated in Table 7 and is a single change: the source is one calendar year, so the fifteen-month training block does not exist and the calendar *position* of the split is preserved instead — validate on April and May, test on June, as in every other row — leaving three months of training rather than fifteen. That biases against us, which is the safe direction to err.

The audit. This is the only tier-2 source whose documentation describes a construction rather than a measurement. Wu and Kan take each day’s maximum and minimum from National Development and Reform Commission reporting, trace a representative workday profile and a representative holiday profile off published load curves with WebPlotDigitizer, and rescale one of those two profiles to each day’s envelope [Wu and Kan, 2023]. Taking that on trust is exactly what this project does not do elsewhere, so we checked it against the data: min-max normalise every calendar day’s 24-hour profile and count the distinct shapes in a year.

The answer is two, in every province. Table 10 reports it: 778 numbers — two normalised day-shapes and 365 (min, max) pairs — reproduce all 8,760 hourly values to within 2×10^{-9} of their magnitude, which is the precision the source is published at rather than an approximation we introduced, at a compression of better than eleven to one. The documentation is accurate.

What that disqualifies. Skill in this study is a ratio against each series’ own seasonal-naive baseline. When every workday shares one shape, seasonal naive recovers that shape exactly and its entire error is the daily envelope — and so is ours. Both models collapse onto predicting the same two administrative numbers per day, the ratio between them collapses toward zero, and the observed mean skill of +0.040 with two negative rows is what that predicts rather than a finding about a forecaster. The China skill column is reported for completeness and is not evidence either way about the method.

This costs us the arm’s most quotable result. Guangdong is the closest structural analogue to Delhi anywhere in the study — cooling share 0.85 against Delhi’s 0.89, load-temperature correlation +0.37 against +0.59, a comparably peaky profile — and it returns -0.314, the second-worst skill in the benchmark. Read naively that corroborates the Indian result on independent 2018 data from another country and retires the objection that Delhi’s weakness is an artefact of a decade-old archive. It does not,

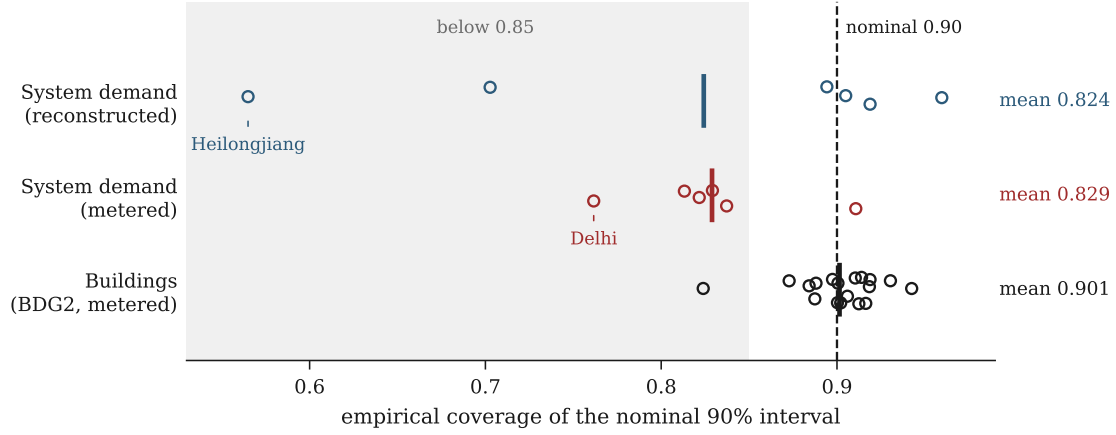


Figure 2: One dot per supply. The nominal 90% interval very nearly holds on individual buildings and fails systematically once demand is aggregated to system level, on both the metered panel and the reconstructed one. The two populations of system demand share no country, no year and no data-generating process, and their means differ by 0.005.

Correlation against forecast skill	r	n	leave-one-out range
<i>Panel A — 18 BDG2 buildings, three countries</i>			
cooling-degree-day share	+0.010	18	-0.308 to +0.077
controllable (HVAC) fraction	+0.283	18	+0.029 to +0.378
mean outdoor temperature	-0.012	18	-0.266 to +0.051
median load	-0.014	18	-0.189 to +0.120
<i>Panel B — 6 Chinese provinces, independent replication</i>			
cooling-degree-day share	+0.054	6	-0.377 to +0.582
mean outdoor temperature	+0.216	6	-0.290 to +0.782
load-temperature correlation	-0.060	6	-0.619 to +0.713
<i>controllable fraction vs climate (not vs skill)</i>	+0.489	18	—

Table 9: Forecast skill is uncorrelated with climate, controllable fraction, temperature and load size, and every leave-one-out range straddles or nearly straddles zero — which at these sample sizes is what a null looks like. Panel B is the replication on a panel the null was not fitted to. The final row is the one relationship in the study that is real, and note what it is between: available flexibility tracks climate strongly, while predictability does not track anything. Skill and flexibility are independent deployment axes.

because the number is a property of the construction. We report the temptation as well as the retraction, because the inference is one a reader would otherwise make on our behalf.

What survives. Coverage is a property of interval width against realised error and does not run through the baseline, so the tier-2 result can be read on this arm — and it replicates: mean 0.824 against 0.829 on the metered panel, on an error process unlike any other in the study, with the worst row anywhere at 0.565. That two populations sharing no country, no year and no data-generating process land within 0.005 of each other is stronger evidence for the aggregation effect than a larger metered sample would have been. The null of Section 7 replicates too. The wider coverage spread on this panel (0.565–0.960 against 0.762–0.911) is consistent with the three-month calibration history and is not separately identified.

What the exercise is worth. A constraint imposed late produced no new capability and one new measurement, which is a characterisation of the dataset rather than of the method, and which removed a result we would have liked to keep. That is the outcome we would want from an audit, and it is reported at the same size as everything else.

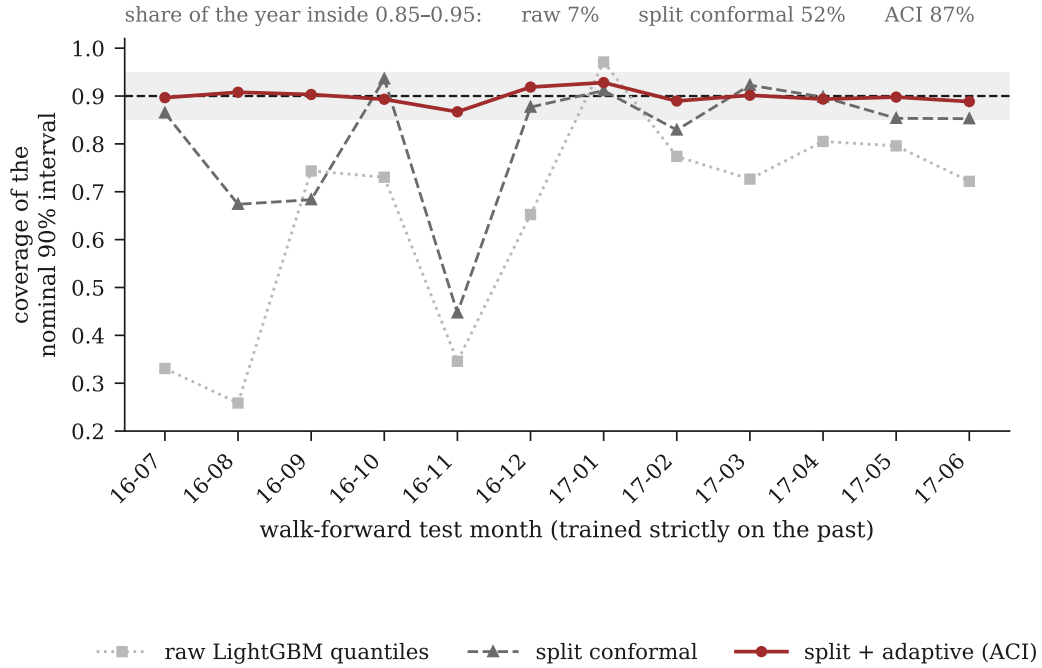


Figure 3: The remedy, measured. Twelve monthly folds, each trained strictly on its own past and calibrated on the thirty days immediately before it. Raw quantiles are not calibrated out of sample at all. Split conformal — the layer with the theorem attached — narrows the failure and does not remove it, bottoming out at 0.447 in a month whose ceiling was being defended against a bound reality broke through a third of the time. Adaptive conformal inference holds 0.867–0.928 in every month of the year.

9 India in comparison

Table 11 is the comparison the benchmark exists to make, and it resolves into a single sentence: **India is the place this method is most worth deploying and the place it is currently least trustworthy.**

Why India is the favourable case. Delhi has the highest cooling-degree-day share of any metered series in the benchmark at 0.89, and it is the only series whose demand rises *strongly* with temperature (+0.59). France is its mirror at -0.64 : French demand rises when it gets cold, because the building stock is electrically heated. That distinction is the whole economic argument. A controller that buys down a cooling peak inside a demand-charge tariff has something to work with in the first regime and almost nothing in the second. Delhi is also among the peakiest profiles in the set ($p99/\text{median}$ 1.61), which is precisely the shape a demand charge punishes.

Why India is the difficult case. On both axes that determine whether the controller can be relied upon, India is last among metered supplies. Skill over a seasonal-naïve baseline is +0.097, against +0.713 for Germany, +0.636 for France and +0.619 for Spain — a model that needs no training at all is nearly as good on Delhi as ours. And coverage of the nominal 90% interval is 0.762, the worst metered figure anywhere in the study, against 0.911 for Ireland.

The second of these is the serious one. The controller’s $q95$ is read from the same object as that interval, so an interval that is too narrow yields a demand ceiling that is too narrow, and the safety property is weakest on exactly the load the project targets.

10 What India must fix

Stated as work rather than as caveats, in descending order of how much they cost.

1. There is no Indian building-level load dataset. This is a national research-infrastructure gap and it bounds everything else. The Building Data Genome Project 2 harmonises 1,636 buildings across three countries and contains no Indian building. We were therefore forced to test the *controller* on

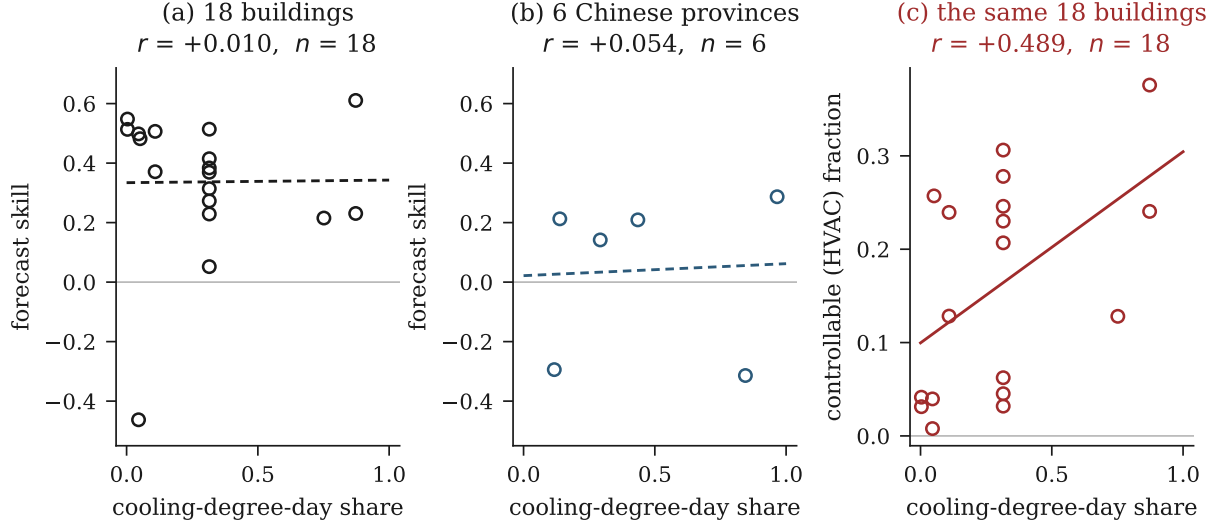


Figure 4: (a) The null. (b) The same null on a panel it was not fitted to, in a different country, year and data-generating process; panels (a) and (b) share a vertical scale. (c) The same eighteen buildings, the same horizontal axis, and the one relationship in the study that is real. How much load there is to shift tracks climate. How well it can be predicted does not.

Province	Days	Shapes	Days/shape	Numbers	Values	Max err. %
Hainan (Haikou)	365	2	115/250	778	8,760	0.000000
Guangdong (Guangzhou)	365	2	115/250	778	8,760	0.000000
Shanghai	365	2	115/250	778	8,760	0.000000
Beijing	365	2	115/250	778	8,760	0.000000
Yunnan (Kunming)	365	2	115/250	778	8,760	0.000000
Heilongjiang (Harbin)	365	2	115/250	778	8,760	0.000000

Table 10: Provenance audit of the China arm, run against the data rather than taken from its documentation. Every calendar day’s 24-hour profile is min–max normalised and the distinct shapes counted. *Shapes* is how many survive; *Numbers* is how many values are needed to rebuild the year from them; *Values* is how many the year contains. A whole year of hourly provincial demand is 778 numbers — 2 normalised day-shapes and one (min, max) pair per day — reproducing all 8,760 values to within $2e-09$ of their magnitude, which is the precision the source is published at rather than an approximation we introduced. Compression 11.26:1. The published method is exactly what the authors state it is. The consequence for this study is that forecast skill, a ratio against a seasonal-naïve baseline, is uninformative on these series: the shape the baseline has to guess is constant, so both models reduce to predicting the same two administrative numbers per day.

American buildings under an Indian tariff and the *forecaster* on Indian city demand, and no single row of this study is both Indian and building-level. Relabelling an American building as Indian would have been worse than the gap, so the gap is reported. Until an Indian equivalent exists — submetered, multi-site, multi-year, openly licensed — no one can honestly claim a building-level demand-control result for India, ourselves included.

2. Calibration collapses across the monsoon. Coverage of 0.762 is an assumption failure, not a tuning failure: split conformal requires calibration and test blocks to be exchangeable [Vovk et al., 2005, Romano et al., 2019], and a Delhi May is not exchangeable with a Delhi June. Section 6 shows that the adaptive layer fixes exactly this failure mode at building scale, holding 0.867–0.928 across a walk-forward year where split conformal falls to 0.447. That is the strongest available reason to expect it to help at system scale, and it is not the same thing as having measured it there, which we have not.

3. Indian load carries calendar structure the model does not encode. A large share of Indian public holidays are lunisolar — Diwali, Holi, Eid — and move against the Gregorian calendar, so they cannot be expressed as the fixed date list that load-forecasting pipelines conventionally use, including ours until this study forced the change. Beyond holidays, Indian demand carries load-shedding events, agricultural pumping schedules and festival periods that have no counterpart in the European series and

Series	Provenance	Native min	Cooling share	Load-temp. corr.	$p99/med$	Skill	Cov 90%
Hainan (Haikou)	recon.	60	0.97	+0.10	1.32	+0.287	0.960
India (Delhi)	metered	15	0.89	+0.59	1.61	+0.097	0.762
Guangdong (Guangzhou)	recon.	60	0.85	+0.37	1.47	-0.314	0.919
Shanghai	recon.	60	0.44	+0.07	1.64	+0.209	0.894
Beijing	recon.	60	0.29	-0.28	1.59	+0.142	0.905
Spain	metered	60	0.28	+0.17	1.31	+0.619	0.813
Yunnan (Kunming)	recon.	60	0.14	-0.35	1.24	+0.213	0.703
Heilongjiang (Harbin)	recon.	60	0.12	-0.47	1.31	-0.294	0.565
France	metered	60	0.08	-0.64	1.58	+0.636	0.829
Germany	metered	60	0.06	-0.10	1.32	+0.713	0.837
United Kingdom	metered	60	0.02	-0.13	1.40	+0.161	0.822
Ireland	metered	60	0.00	-0.07	1.37	+0.442	0.911

Table 11: India against every other system-level series in the study, ordered by cooling-degree-day share. Among metered supplies Delhi is the extreme of both halves of the verdict: the highest cooling share, the strongest positive load-temperature relationship, the peakiest profile, and simultaneously the lowest forecast skill and the worst interval coverage. The six Chinese provinces, added after the study was written, displace India from three of those four extremes on the face of the table — and the provenance column is why that displacement is reported rather than acted on. Guangdong is the closest structural analogue to Delhi in the study and returns the second-worst skill in it, which reads like corroboration and is not: Section 8 shows the Chinese skill column is an artefact of how the data was built. Delhi remains the only series at native 15-minute resolution, the cadence the controller runs at and the cadence Indian demand charges are assessed on.

no representation in our feature set. The low skill figure of +0.097 should be read partly as a statement about missing features rather than solely about the data.

4. The Indian series is a decade old. The Delhi archive covers 2011–2012. India’s generation mix, air-conditioning penetration and peak demand have all moved substantially since. The load *shape* conclusions are the durable ones; the levels are not, and nothing in this paper should be read as a statement about the present Indian grid. We had a candidate answer to this objection in the China arm and Section 8 withdrew it.

What this does not undermine. The favourable half of Table 11 rests on physical structure — cooling share, the sign of the load-temperature relationship, peakiness — which is the part least likely to have inverted since 2012. Indian demand has not stopped rising with temperature.

11 Results that argue against the method

Five rows are reported unchanged, and three of them are outright negative — worse than a baseline that needs no training at all.

The London office returns skill -0.463 and public services in Washington DC returns $+0.052$. Both sit at 1–3% HVAC share, where a weather-aware model has little to add over “last week, same time” — a coherent and testable explanation rather than an excuse. Delhi returns $+0.097$ with the study’s worst metered calibration, and its explanation is different and given in Section 10: a system-level series carrying calendar structure the feature set does not encode. Guangdong and Heilongjiang return -0.314 and -0.294 , the two lowest figures in the study, and Section 8 explains why those two should not be used against the method any more than they should have been used for it.

Two defects in our own pipeline were exposed by running it outside its home site. A hard-coded United States federal holiday calendar would have inverted the holiday flag on two of the largest annual load anomalies for every European building; holidays are now resolved per country, and India is the case that forces it, its calendar being substantially lunisolar and therefore not expressible as a fixed list. And a data-preparation stage overwrote its own manifest, silently discarding every site but the last, which single-site work had never exposed.

A third is worth recording because it is the kind that survives review. The column commonly written as “the Boole bound” in applied chance-constraint work is usually $1 - (1 - \alpha)^H$, which is the independence calculation and is not a bound. Our own tables carried that label until this draft. The corrected reading in Section 4 is stronger than the one it replaces, because the valid bound turns out to be vacuous over the horizon that matters.

12 Limitations

The system tier is a forecasting benchmark, not a control benchmark: there is no thermal mass to shift at grid level and no savings claim is made from it. The Delhi window (2011–12) does not overlap the European one (2016–17) or the Chinese one (2018), which motivates reporting within-series skill rather than absolute loss but does not eliminate the concern. Capital-city temperature proxies national load-weighted temperature, acceptable for Great Britain and Ireland and rough for Germany, France, Spain and every Chinese province. Nothing here claims that model *weights* transfer across countries — our own cold-start study finds the model losing to seasonal naive on an unseen building at the same site, scoring 3.546 pinball against that baseline’s 3.037 — only that the architecture and protocol do. The demographic arm is a single city and cannot separate demographic from local building stock. Delhi is one Indian city, and is real Indian data rather than a claim about India. The China arm is reconstructed data and Section 8 states exactly which of its columns may be read. The horizon and acceptance results are one building over one month; they are the deepest measurements in the study and the narrowest.

13 Conclusion

We built an instrument rather than an algorithm and ran it unchanged over thirty electricity supplies in eight countries to find out where a tariff-native demand controller’s guarantees actually hold.

The central result is that the guarantee most commonly deployed is not the guarantee most commonly claimed. A per-step chance constraint on a demand ceiling, marginally calibrated and behaving exactly as specified, is under-protected over a sixteen-hour planning window by a factor of eight; the figure usually quoted as the conservative correction is not a bound and overstates the risk by $2.4\times$; and the correction that is a bound is vacuous four hours in. That risk level can be turned into an operator-set parameter, at a measured price of Rs 3,339 on a billing month and with its inexactness and resolution floor stated. Alongside it, distribution-free coverage degrades with aggregation level and it is the adaptive layer rather than the one with the theorem that holds it; and forecast skill and available flexibility are independent axes, only one of which tracks climate.

For India the benchmark returns a divided verdict, and both halves matter. India has the most favourable load physics in the study — the highest cooling share, the only strongly positive load–temperature relationship, a peaky profile that a demand-charge tariff punishes and this method is built to defend. It also returns the lowest metered forecast skill and the worst metered calibration we measured, and there exists no Indian building-level dataset on which to close the loop properly. The first half is the case for the work. The second half is the work that remains, and Section 10 states it as such rather than as a footnote.

Every number here is regenerated by one script from a clean checkout. The results that contradict us are printed at the same size as the rest, and the one result we most wanted to keep is the one Section 8 withdraws.

Reproducibility

```
python data/national.py          # fetch and cache every system-level series
python eval/comparative.py       # run the protocol over all thirty supplies
python eval/china_audit.py       # audit the China arm against its own data
python eval/horizon_risk.py      # the bracket, the copula and the closed loop
python eval/conformal_audit.py   # the walk-forward calibration year
python eval/paper_tables.py      # emit every table in this paper
python eval/paper_figures.py     # emit every figure in this paper
```

References

- Quentin Berthet, Mathieu Blondel, Olivier Teboul, Marco Cuturi, Jean-Philippe Vert, and Francis Bach. Learning with differentiable perturbed optimizers. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- Giuseppe C. Calafiore and Marco C. Campi. The scenario approach to robust control design. *IEEE Transactions on Automatic Control*, 51(5):742–753, 2006.

- Priya L. Donti, Brandon Amos, and J. Zico Kolter. Task-based end-to-end model learning in stochastic optimization. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- Adam N. Elmachtoub and Paul Grigas. Smart “predict, then optimize”. *Management Science*, 68(1): 9–26, 2022.
- Isaac Gibbs and Emmanuel J. Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- Tilman Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.
- Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. Lightgbm: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- Clayton Miller, Anjukan Kathirgamanathan, Bianca Picchetti, Pandarasamy Arjunan, June Young Park, Zoltan Nagy, Paul Raftery, Brodie W. Hobson, Zixiao Shi, and Forrest Meggers. The building data genome project 2: energy meter data from the ashrae great energy predictor iii competition. *Scientific Data*, 7(1):368, 2020.
- Arkadi Nemirovski and Alexander Shapiro. Convex approximations of chance constrained programs. *SIAM Journal on Optimization*, 17(4):969–996, 2006.
- Kingsley Nweye, Siva Sankaranarayanan, and Zoltan Nagy. Citylearn: A tutorial on reinforcement learning control for grid-interactive efficient buildings and communities. In *Proceedings of the ACM International Conference on Systems for Energy-Efficient Built Environments (BuildSys)*, 2023.
- Yaniv Romano, Evan Patterson, and Emmanuel J. Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.
- Haochi Wu and Xiaoming Kan. Hourly electric power load and transmission data at the provincial level in china, 2023. Dataset. Hourly load for 31 provinces, 2018, reconstructed from NDRC daily maxima and minima and digitised representative load curves.